How TabPFN-TS makes a forecast

Prior Labs · TabPFN-v2 regressor · a tabular transformer used for forecasting · zero-shot

1. One series, plus covariates

TabPFN-TS reads one series and any covariates known ahead. A ? is a value not known yet.

past *future*

Target temperature

Known ahead day length

Past-only covariates are not used; five metals become five separate tables.

2. Turn time into a table

Each step becomes a row: a running index, calendar sine and cosine terms, detected seasonal terms and the covariates. No lagged values.

index	sin	cos	cov.	y
				?
				?

Rows with y known are the training rows; the horizon rows have y = ?.

3. Twelve two-way attention layers

The whole table goes in at once; the training set is just more input (in-context learning).

Across features: each cell reads the other cells in its row

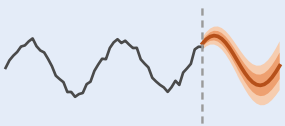
Across rows: each row reads the training rows; horizon rows never read each other

Feed forward after each attention layer

Pre-trained only on synthetic tables, never on real time series.

4. A distribution for each row

For each horizon row the output is a histogram over y; its quantiles and median are read off.



median

60% band

80% band

Several passes with shuffled preprocessing are averaged (2 on a CPU in our runs).

No forecasting architecture at all: a table model predicts the horizon rows from the past rows.

12 layers · attention across features and across rows · output: a discretised distribution per row

In the notebooks: ungated TabPFN-v2 weights · last 2,048 days or full monthly history · calendar covariates for D2 · median as the point